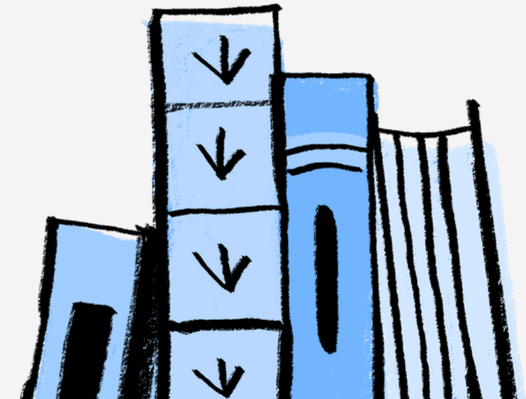






SCIENTIFIC WRITING + RESPONSIBLE AI USAGE





Today's Agenda

01 Basic Guidelines

02 Citations + Sourcing

03 Introduction Parts

04 Methods Parts

05 Results Parts

06 Conclusions Parts

07 Responsible AI Usage Talk

08 Fun Activity!

Guidelines of Scientific Writing

Be clear and concise: Use plain language + Dont add filler

Use appropriate tenses: Use the **past tense to describe your methodology and findings**, and the present tense for established facts, general truths, and conclusions.

Write for your audience: Read recent papers in your target journal to match the publication's style and terminology expectations

Citations & Sourcing

There are different conventions for scientific writing depending on journal, but most accept a references page at the end of your article, with numbers and full citations

- For in-text citations, include numbers and bracket (varies, but this is the convention fora lot of larger journals)
- Ex: [1...3]
  - [1,5]
  - [6]
- (Numbers correspond to order listed in References Page. Normally numbered in order of usage; first paragraphs have 1-6, next have 7-13, etc.)

Citations & Sourcing

References

[1] G. Hanaoka, Y. Sakai, T. Shimizu, T. Shimoyama, and S. H. Shin, "A setup-free threshold encryption scheme for the bitcoin protocol and its applications," *IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences*, vol. E103A, no. 1, pp. 150–164, 2020, doi: 10.1587/transfun.2019CIP0018.

[2] L. M. Lach *et al.*, "The health and psychosocial functioning of caregivers of children with neurodevelopmental disorders," *Disability and Rehabilitation: An International, Multidisciplinary Journal*, vol. 31, no. 9, pp. 741–752, 2009, doi: 10.1080/0891693080235494 8.

[3] E. Zaghloul, T.T. Tong, M. W. Mutka, and J. Ren, "Bitcoin and blockchain: Security and privacy," *IEEE Internet Things J.*, vol. 7, no. 10, pp. 10288–313, Oct. 2020, doi: 10.1109/JIOT.2020.3004273.

[4] T.H. Chang and D. Svetinovic, "Improving bitcoin ownership identification using transaction patterns analysis," *IEEE Trans. Syst. Man Cybern., Syst.*, vol. 50, no. 1, pp. 9–20, Jan. 2020, doi: 10.1109/TSMC.2018.2867497.



Writing the Introduction: Parts of an Intro Section

1: HOOK AND BACKGROUND

Goal:

Grab the reader's attention and explain why the topic matters globally or within your field.

Include:

Define key terms (which you should have kept track of on your own!)

Read back and find for what terms may not be commonly known

2: THE LITERATURE REVIEW (IT'S BACK!)

Goal: Show the reader what is already known and establish credibility

Include: A brief summary of key, recent, and highly relevant studies.

Treat your sources as a **collection!**

3: RESEARCH GAPS + YOUR WHY

Goal: Justify why your paper is needed

Include:

A specific limitation, contradiction, or missing piece in current research.

Example: "Although previous studies have examined [Topic X], Topic Y remains understudied"

4: OBJECTIVE

Goal: Tell the reader exactly what your study will do.

Include:

- Your primary research question, hypothesis, main objectives, and a brief overview of how the rest of the paper is organized

Example Sentence: “The present study aims to ___.”

“We (if a team or +1 contributors)/I hypothesized that ___”

ADDITIONAL NOTES:

- This section's most important role is to answer what did other people miss that you want to now answer?
- This should be a large chunk of writing, generally spanning from 1-2 full pages
- If you're having trouble writing this, find a basic article (maybe the workshop article from a few weeks ago!), and notice the format



Writing the Materials/Methods: Organizing Methods in a Paper

Materials + Methods

- Most research papers will split this up into **sections chronologically** to make the paper easy to read and **replicate**.

- Some of these "**subsections**" are given names specifically to the study, though here we will give the general idea for hese subsections

1: STUDY DESIGN

Describes overarching research framework

Examples:

- randomized controlled trial
- longitudinal study
- computational project
- in-vitro (takes place in controlled lab setting, outside of a living organism)
- in-vivo (in a living organism)

Format:

Since Materials and Methods is the 2nd section after Introduction, you will write each subsection as 2.1, 2.2, 2.3, etc.

Example:

2. Materials and Methods

2.1 Study Design (this is italicized)

The present study was utlized ___.

2: PARTICIPANTS, SUBJECTS, OR MATERIALS

Example title:

- *2.2 Participants*
- *2.2 Material Collection*

- Include ALL the specific information someone would need to know
- Have inclusion/exclusion criteria
- **Ex**: "We surveyed 50 students from Katherine Johnson Middle School about whether they played sports, but 5 of them had an injury so they were excluded from the study (n=45)."

3: PROCEDURES AND DATA ANALYSIS (STATS)

Procedures: A chronological account of the experiments or data collection methods

- *2.3 Procedures and Data Analysis*

Data Analysis (Statistics): Outlines statistical tests & programs used to evaluate the data

"A simple linear regression was used to relate x and y"
"We used Stapplet to visualize the descriptive statistics of our data"



Writing the Results: What to Include and What to Avoid

Results: What to Include

- **Data Summary**: Present data using text, tables, and figures **without interpreting the findings**
- **Statistical Outcomes**: Report both significant and non-significant null results alongside appropriate test values, degrees of freedom, and exact p-values
- Connect results to research questions from introduction

Results: What to Include EXAMPLE

- **Restating** what you did:
  - Ex: "To investigate the time exercised of a middle schooler by grade, we conducted an xyz survey. The survey revealed that middle schoolers in grade 6 exercised significantly greater than in grade 8 (p=0.002)"

- **Subection may be**: *3.1 Exercise decreases across grades*

Results: What to AVOID

- **Avoid Interpretation:** Don't explain results (yet!)
- Avoid unanalyzed + raw data → SUMMARIZE DATA
- Mix the order of methods and results



Writing the Discussion

Discussion

- **Interpret your results**
  - Give a possible answer for WHY something occured
  - “This result SUGGESTS ____”
- **Compare them with existing literature**
  - How does your result agree/disagree with others?
- **Acknowledge study limitations**

What are Agents?

- Agents are a smart helper that can think, decide, and do things to achieve a goal

- Ex. making an app: one agent builds front-end, one agent setsup database, one tests code

Claude

GPT-5 Codex

Use AI to GROW, not STUNT your learning

Fun activity: Haiku's Survey!

THANK YOU!